Section A: Data Fundamentals

1. Database is a shared, integrated computer structure that stores the collection of end-user data and metadata
- Databases make data persistent and shareable in a more secure way.
- In businesses and social media.

2. Central repository used to store valuable information and organize large volumes of current and historical data, that is structured
- Data warehouse is an Online Analytical Processing used for data analysis in order to make business decisions while database is Online transactional processing that enables real-time execution.
- Company makes use of a data warehouse for analysis and strategic business decision.

3. Data Lake a central repository that stored unstructured, semi-structured and structured ~~and unstructured data~~ in its raw format. Allows for processing in its original form. Mainly for ~~data~~ machine learning.
- Preferred when all the data forms are stored and must be used regardless that it is processed

4 Data Mart is subject-oriented repository designed to serve specific analytical needs for a specific department, Making smaller Data warehouse is large, central repository.

5. Unstructured data - data in its raw form. A video
- Semi-structured data - Structured to some extent, ~~a python file~~. A JSON file
- Structured data - data processed to ~~e~~ reveal its meaning, building block to information. Data in databases.

6. Designing how data is structured, stored and related. ~~Make~~ Ensures data is organised for increased end-user productivity, accuracy, validity and timeliness.
- Conceptual
- Logical
- Physical

} data model

7. Analyzing data to find patterns, trends and useful information. Used to understand customer behavior.
- Retail
- Banking

8. Intermediary between end-user and database. Enables data to be shared.
- MYSQL: Open-source relational DBMS
- OracleDatabase: Commercial relational/ DBMS
- MongDB - ~~Enterprise systems Move~~ App development
- PostgreSQL - Data-intensive applications

9. Process used to collect data from different sources, clean and prepare it, and store it in a system for analysis
- Extract: Data is collected from various sources.
- Transform: Data is cleaned
- Load: Data is loaded into a target system
- Enable data warehousing.

10. Comma-Separated Values
A simple plain text format where each line represent a record & filed are seperated by a delimiter.
Apache Avro – A compact binary row-based storage format. It stores data with itself allowing for a robust "scheme evolution)

Section B: AI and ML

11. Computer system that imitate human intelligence used to help make decisions.
- Weak AI: Designed to perform one specific task.
- Strong AI: Machines that can perform any intellectual task that a human can do.

12. ~~Mach~~ Subset of AI that enables systems to learn from data. Training / Learning purposes. Does does not rely on manualy coded rules, they are automatic.
Supervised - need human interaction.
Unsupervised - can make and take decision without human interaction.
Reinforcement Learning – Agents learn make to make decisions by trial and error, interacting with an environment to maximise cumulative rewards.

13. Teach computers to learn complex patterns automatically. It is subset of Machiene Learning. Network of layers.

14. Enables computers to understand, interpret and respond to human language (spoken or written)
- Translation
- Autocorrect
- Voice assistant

15. LLM are advanced AI systems built on deep neural networks designed to process, understand and generate human-like text.
- Standard NL model is trained on smaller, structured databases and LLM is trained on huge amounts of unstructured text.
  - GPT-4: Advanced multimodal reasoning and capabilities.
  - LLaMA3: Open-weight model, popular in research and startups.

16. Generative AI is a subset of ai focused on creating new content rather than just analyzing or classifying existing data.
- ChatGPT: Generates human-like text based on prompts.
- DALL-E: Generates images from text description
- GitHub Copilot: Generates code suggestions while you are programming.

17. Training dataset is the data used to teach a machine learning model. Because of the GIGO rule, Traing a dataset using baised data will result in baised model. Biased decision making, unemployement and inequality. Models that make decisions with no to little human.

18. A field of ai that enables machines to 'see' process and understand images or videos. They process imgaes as data. The image is made up of pixels and each pixel has numerical values.
- facial recognition.
- self-driving cars.

Section C: APIs, Tools & Developer Concepts.

19.- A set of rules or protocols that enables software applications to communicate withe each other.
- Computer in the same Local Area Network communicating with each simultaneously, to be able to know which computer is communicating with it makes use of API address.

20. An API key is a unique code used to identify and authenticate a user or application when accessing an API. Its used for ~~aut~~ authentication, access control and usage tracking.
- Can make use of third parties, or regenerate keys regularly.
- It will lead to data breaches, financial loss and unathorized access.

21. REST APIs communicate through HTTP requests to perform standard database functions.
→ REST : Representational State Transfer
→ Get : Used to retrieve data from the server
→ POST : Used to create a new resource
→ PUT : Used to update or replace an existing resource entirely.

22. JSON is a lightweight data format used to store and exchange data between systems.
- Easy to read and write and Ideal for APIs.

```
{ "EMPLOYEE": 0002,
  "Name": "Sam",
  "Age": 27;
  "Salary": R70 000;
}
```

23. Practice of designing inputs to language models that reliably produce the outputs you need.
Directly influences the quality, relevance and accuracy of generative AI outputs.

23. ~~Prompt Engineering.~~

24. Map function is a function applied to each element in a collection and produce a new collection of results. Transform each item in a list using a rule.

e.g Convert a list of string into a list of integers

```
String = ['1','2','3','4'];
res = map(int, string);
print(list(res));
```

Output : [1, 2, 3, 4];

25. Delivery of IT resources over the internet with pay-as-you-go pricing, allowing users to rent infrastructure and software rather than owning physical data centers.

- Iaas → cloud computing that delivers on-demand access to cloud-hosted compute, storage and networking
- Paas → cloud computing model that provides a complete on demand cloud platform for developing, running, maintaining and managing applications
- Saas → cloud-hosted application software delivered over the internet to compatible computing devices
- ~~Storage (Iaas)~~ Website hosting (Iaas)
- API development and management (Paas)
- Google Docs (Saas)

26. A tool used in a software development and collaborative projects to track and ~~manag~~ manage changes to source code.

- Git is a distributed version control system, gives every used a complete local copy of the entire project
- Commit: A snapshot of your changes at a specific point in time. When you "commit" code, you save it permanently into a project's history with a descriptive message.
- Branch: An independent line of development that diverges from the main codebase. ~~It allows you~~ It allows you to make changes to be made in an isolated environment without effecting the main branch until you are ready to merge the changes back.

27. ~~Jupyty~~ Jupyter Notebook is an ~~a~~ interactive environment that ~~lets~~ lets you write code, run it and display results in one document. Most commonly used with Python

28. Python is one of the most popular programming language, ~~[illegible]~~ Often used to build websites and software, automate tasks and conduct data analysis. Ease of use, strong community support.

- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Section D: Data Ethics, Governance & Careers.

29. refers to the protection of personal information. Protect personal identity, prevent misuse of information & others. GDPR is Europe's new data privacy & security law. Applies to EEA (European Economic area countries

30. refers to principles & guidelines that govern how data should be collected, used, shared & managed in a fair, respectful & respectful manner. Misuse of personal data, Deepfakes & AI manipulation & Biased in AI hiring

- Violates privacy rights
- leads to unfair discrimination
- Spread false information.

31. is the data management discipline that focuses on the quality, security & availability of an organization's data.

- Data security and privacy & data quality management.

32. is a phenomenon where artificial intelligence system produce systematically skewed, unfair or discriminatory results often reflecting prejudice or flawed data.

33.

- Data engineer develops and maintains data ~~Engineer~~ architecture and pipelines.
  - They build programs that generate data, data meaningful for operations and analysis.

- Data scientis studies large data sets using advanced statistical analysis and machine learning algorithms.
  - They identify patterns in data to drive business insights.

- Data analyst extract insights from data to help businesses make better decisions.
  - They identify trends and patterns from the data to make insightful decisions

~~Data~~ E	Data Engineer	Data Scientist	Data Analyst
Role	- Optimize data warehouse - Build and maintain data platform - Manage pipline ~~orchestration~~ orchestration	- Apply data techniques to gaing insights - Identify patterns & trends - Develop ~~orchestration~~ machine learning algorithm	- Collect & clean data - Data analysis and interpretation - Data visualization and reporting
Skills	- ~~Ex~~ SQL - Programming - Experience in engneering and testing tools.	- Creative thinking and decision - making abilities - Experience in big data tools	- Programming - Data visualization tools - Communication & presentation skills

35. Computer Vision. Computer seen images as numeric data. The fact that self-driving cars are able to precisely drive around and determine the activities in the surrounding areas based off numeric data. Thats ~~mary~~ mind blowing.

3.6 API. Still dont get it, ~~that~~

37 Data and AI do more good than harm. People at the fore front of technology and innovation hold much power to do train the models based on their motives and unfortunately run the technology world.

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